

Background

Our organization receives incident notifications from multiple channels: Slack, email, phone calls, and WhatsApp. Customers expect to simply send an email or a WhatsApp message when something is wrong, without having to use overly complex tools or forms.

However, many incident management work with complicated forms or expect customers to select a severity level themselves (so what ends up happening: everything is extremely severe, even if it's a small bug). This is impractical and often inconsistent. We want to remove this burden: instead of asking the reporter to decide, the system should automatically determine severity based on the content of the message. For example, if a customer reports downtime, the system should automatically mark the incident as Critical.

We are currently in the process of looking into ways that we can automate our incident management, to avoid being dependent on a single project manager or developer. So we are hoping to build a solutions that integrates, slack, email and ideally WhatsApp together and is able to automatically assign through a slackbot incident commanders and get the process started.

Problem Statement

Currently, severity assignment is manual and depends on human judgment. This leads to delays, inconsistency, and slower response times. We need a solution that:

- Integrates directly with Slack, email, and ideally WhatsApp,
- Automatically captures incoming messages
- Classifies severity levels automatically
- Assigns an incident commander based on the person assigned as incident commander that week or with higher severity levels follows a set of rules

Goals of the Assignment

The student team should design and implement a standalone proof-of-concept system that:

1. Integrates with Slack to process incident messages.
2. Captures incoming email and WhatsApp messages (e.g., via APIs or a bot) and converts them into structured incident reports.
3. Automatically assign severity levels (Minor, Major, Critical).

4. Ensures the system is extensible, so future integrations (e.g., other monitoring tools) can be added.
5. Suggests or assigns an Incident Lead automatically, using a round-robin method among available developers.

Deliverables

- A working prototype that demonstrates:
 - Slack, email, and WhatsApp message → Processed by system → Automatic severity assignment
- Documentation describing the architecture, configuration, and severity rules.
- A final presentation/demo to stakeholders at the end of the project.

Skills & Technologies

- API integrations
- Backend development (Java, springboot)
- Docker
- Angular frontend, with a suggestion to use tailwind